



## Statistics of Financial Markets

### Chapter 7: Efficient markets, random walk and variance-ratio tests



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## Acronyms used in this material (1/2)

- ▣ ACF: Autocorrelation Function
- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AMH: Adaptive Markets Hypothesis
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ CME: Chicago Mercantile Exchange
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ EMH: Efficient Market Hypothesis
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ EU: European Union
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GMM: Generalized Method of Moments
- ▣ KPSS: Kwiatkowski–Phillips–Schmidt–Shin (stationarity test)
- ▣ LB: Ljung–Box (test)
- ▣ MA: Moving Average (model)
- ▣ MIT: Massachusetts Institute of Technology
- ▣ OLS: Ordinary Least Squares
- ▣ PP: Phillips–Perron (unit-root test)

## Acronyms used in this material (2/2)

- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ RW1: Random Walk 1 (i.i.d. increments)
- ▣ RW2: Random Walk 2 (independent increments)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ SFE: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- ▣ VR: Variance Ratio
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

## Today's Question and Route

- **Question:** can yesterday's prices tell us anything about tomorrow's return?
  - ▶ if yes, a simple rule could beat the market; if no, the market is efficient in the weak sense
  - ▶ Chapter 4 defined the random walk and the martingale; here we test them on real prices
- **Route** of the chapter
  - ▶ efficient markets: three forms, the joint-hypothesis problem
  - ▶ random walk hypotheses RW1, RW2, RW3 and the martingale
  - ▶ white noise and the autocorrelation function; Ljung–Box and runs tests
  - ▶ unit roots: ADF, Phillips–Perron, KPSS; prices against returns
  - ▶ variance-ratio tests; efficiency across markets and over time; anomalies

## Learning Outcomes

- ▣ State the three forms of market efficiency and explain why every test is a joint test
- ▣ Tell RW1, RW2, RW3 and the martingale apart, and say which of them volatility clustering violates
- ▣ Read a sample ACF with the right confidence band and compute Box–Pierce, Ljung–Box and runs statistics
- ▣ Run and combine ADF, Phillips–Perron and KPSS tests on prices and on returns
- ▣ Compute a variance ratio and its robust test, for one horizon and for several horizons together
- ▣ Judge, on real data, which markets look efficient, and whether this changes over time

## Reading and Tools

- ▣ Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 11 (11.3 expectations and efficient markets, 11.5 random walk tests, 11.6 unit root tests)
  - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 11
- ▣ Econometrics of predictability: [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 2 (random walks, variance ratios) and Ch. 4 (event studies)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch\\_07](#)
  - ▶ ported from the SFE Quantlets SFEtimewn, SFEacfar1, SFEacfar2, SFEacfma1 and SFEacfma2
- ▣ Lecture notebook: [open in Google Colab](#)
- ▣ Video course:  [Efficiency of cryptocurrency markets – A GMM-based analysis](#)

## Can You Beat the Market?

- If past prices predicted future returns, a simple rule would earn money with no risk
  - ▶ traders would use the rule; their trades would move prices until the pattern disappears
- The evidence on professional investors
  - ▶ mutual funds, on average, did not beat a buy-and-hold portfolio after costs [Jensen \(1968\)](#)
  - ▶ few funds show skill large enough to cover their costs [Fama & French \(2010\)](#)
- A survey of the arguments for and against: [Malkiel \(2003\)](#)



The trading floor of the New York Stock Exchange, 1963

Photo: [Thomas J. O'Halloran \(1963\)](#), [Library of Congress](#);  
public domain; [Wikimedia Commons](#)

## 1900: Prices as a Random Walk

- Louis Bachelier, *Théorie de la spéculation* [Bachelier \(1900\)](#)
  - ▶ the first mathematical model of prices: the change of a price is a sum of many small random shocks
  - ▶ the expected gain of the speculator is zero
- [Kendall \(1953\)](#): 22 British price series behave like a “wandering” series
  - ▶ the changes from one week to the next look like independent draws
- [Roberts \(1959\)](#): simulated random walks show “patterns” that chartists see in real prices

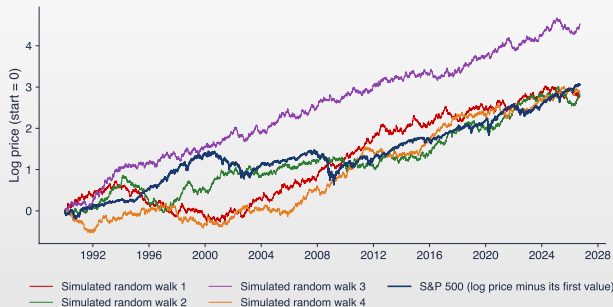


Louis Bachelier (1870–1946)

Photo: [unknown author](#); [public domain](#);

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## Which One Is the S&P 500?



- Four random walks with i.i.d. Normal steps (mean 0.033%, standard deviation 1.13% per day, as the S&P 500) and the S&P 500 log price since 1990
- Question for the room:** without the legend, could you tell which path is real?
- Answer:** hardly; booms and long falls appear in pure noise too (end values: 2.76 to 4.50; S&P 500: 3.06)

## 1965: Properly Anticipated Prices Fluctuate Randomly

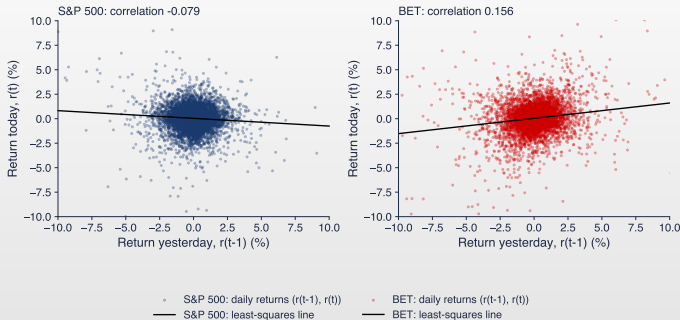
- Paul Samuelson [Samuelson \(1965\)](#): if prices already contain all that traders expect, the next change is unpredictable
  - ▶ formally: the price is a **martingale**,  $E[P_{t+1} | \mathcal{F}_t] = P_t$  ([Chapter 4](#))
  - ▶  $\mathcal{F}_t$ : the information available at time  $t$
- Randomness of price changes is a sign of a market that works, not of chaos
- [Fama \(1965\)](#): daily returns of the 30 Dow Jones stocks show almost no autocorrelation



Paul A. Samuelson (1915–2009), Nobel Prize 1970

Photo: [Bernard Gotfryd \(1970–1975\)](#); public domain;  
[Wikimedia Commons](#)

## Does Yesterday Predict Today?



- Each point: (return yesterday, return today); the line: least squares; its slope is the first autocorrelation  $\hat{\rho}(1)$
- S&P 500:  $\hat{\rho}(1) = -0.079$  (a slight reversal); BET:  $\hat{\rho}(1) = 0.156$  (a slight continuation); are these numbers different from zero?

## Recap: Can Prices Be Predicted?

- From Bachelier (1900) to Samuelson (1965): price changes look random, and economics explains why
- Professional investors rarely beat the market after costs
- Daily autocorrelations are small but not zero: we need tests, not impressions

## What Is an Efficient Market?

- **EMH** (efficient market hypothesis) **Fama (1970)**: prices “fully reflect” the available information  $\Omega_t$ 
  - ▶ no trading rule based on  $\Omega_t$  earns more than the return that compensates its risk
  - ▶ prices move only on **news**: information that was not in  $\Omega_t$
- Efficiency is relative to an information set
  - ▶ the smaller  $\Omega_t$ , the weaker the claim and the easier the test
- Efficient does not mean “returns are zero”: the expected return pays for risk



Eugene F. Fama, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia

[Commons](#)

## Three Forms of Efficiency

| Form        | Information set $\Omega_t$    | What cannot earn excess returns     | Typical test                          |
|-------------|-------------------------------|-------------------------------------|---------------------------------------|
| Weak        | past prices and returns       | technical analysis, chart patterns  | autocorrelation, runs, variance ratio |
| Semi-strong | all public information        | fundamental analysis of public data | event studies                         |
| Strong      | all information, also private | insider information                 | returns of insiders and fund managers |

- The forms are nested: strong  $\Rightarrow$  semi-strong  $\Rightarrow$  weak; a rejection of the weak form rejects all three
- This chapter tests mainly the weak form: past prices are free data for everyone [Fama \(1970\)](#), [Fama \(1991\)](#)

## The Fair-Game Model

- Let  $r_{t+1}$  be the return of an asset and  $E[r_{t+1} | \Omega_t]$  its **equilibrium expected return** (the return that pays for its risk)
  - ▶ the **excess return** (surprise):  $z_{t+1} = r_{t+1} - E[r_{t+1} | \Omega_t]$
- **Fair game**:  $E[z_{t+1} | \Omega_t] = 0$ 
  - ▶ forecast errors are zero on average whatever we know at time  $t$
  - ▶ a rule that invests  $w(\Omega_t)$  earns  $E[w(\Omega_t) z_{t+1}] = 0$  in excess of equilibrium
- With a constant expected return  $\mu$ :  $E[r_{t+1} | \Omega_t] = \mu$ ; returns minus  $\mu$  are a **martingale difference**
- Nothing here says the variance is constant or the distribution is Normal

## The Joint-Hypothesis Problem

- To test efficiency we must know the equilibrium expected return  $E[r_{t+1} | \Omega_t]$ 
  - ▶ this needs a model of equilibrium: constant mean, CAPM (capital asset pricing model), factor models
- Every test of efficiency is a **joint test** of efficiency and of the model [Fama \(1991\)](#)
  - ▶ a rejection may mean an inefficient market, or a wrong model of expected returns
  - ▶ efficiency alone can never be rejected
- For daily returns the problem is small: daily expected returns are tiny (about 0.033% for the S&P 500), so any reasonable model gives almost the same answer

## The Grossman–Stiglitz Paradox

- Information is costly to collect and to analyse [Grossman & Stiglitz \(1980\)](#)
  - ▶ if prices reflected all information perfectly, nobody would be paid for collecting it
  - ▶ then nobody would collect it, and prices could not reflect it
- Conclusion: a market can only be **nearly** efficient
  - ▶ small inefficiencies pay informed traders for their costs
  - ▶ the question becomes “how efficient, where and when?”, not “yes or no?”
- This view leads to the adaptive markets hypothesis of Section 8

## 2013: One Nobel Prize, Two Views



Eugene F. Fama: prices are hard to predict over days and weeks

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons



Robert J. Shiller: prices swing too much compared with fundamentals

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- Both are right, at different horizons: short-run returns are almost unpredictable; long-run returns are partly predictable from valuation ratios

## Recap: Efficient Markets

- Efficiency is defined relative to an information set: weak, semi-strong, strong
- Fair game: excess returns are unpredictable given  $\Omega_t$
- Every test is a joint test with a model of expected returns
- Costly information makes perfect efficiency impossible Grossman & Stiglitz (1980)

## The Random Walk Model of Log Prices

- Log price  $p_t = \ln P_t$ ; **random walk with drift**  $\mu$ :  $p_t = \mu + p_{t-1} + \varepsilon_t$ 
  - ▶ the return  $r_t = p_t - p_{t-1} = \mu + \varepsilon_t$ : the increments  $\varepsilon_t$  have mean 0
  - ▶ by recursion:  $p_t = p_0 + \mu t + \sum_{s=1}^t \varepsilon_s$ : every shock is **permanent**
- If the  $\varepsilon_t$  are uncorrelated with variance  $\sigma^2$ :
  - ▶  $E[p_t] = p_0 + \mu t$  and  $\text{Var}(p_t - p_0) = t\sigma^2$ : the uncertainty grows with the horizon
  - ▶ the  $q$ -day return  $r_t(q) = r_t + \dots + r_{t-q+1}$  has  $\text{Var}(r_t(q)) = q\sigma^2$
- Log prices, not prices:  $P_t$  cannot become negative, and log returns add over time ([Chapter 1](#))

## Three Random Walk Hypotheses

- ▣ **RW1** (i.i.d. increments):  $\varepsilon_t$  independent and identically distributed,  $\varepsilon_t \sim \text{IID}(0, \sigma^2)$  [Campbell, Lo & MacKinlay \(1997\)](#)
  - ▶ the strongest: no function of the past predicts any function of the future
- ▣ **RW2** (independent increments):  $\varepsilon_t$  independent, but not identically distributed
  - ▶ allows the variance to change over time in a way that is not predictable from the past (regimes)
- ▣ **RW3** (uncorrelated increments):  $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$  for all  $k \neq 0$ 
  - ▶ the weakest: allows  $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$ , that is, volatility clustering
- ▣ Nested:  $\text{RW1} \Rightarrow \text{RW2} \Rightarrow \text{RW3}$ ; tests of RW3 use only autocorrelations of returns

## What Each Hypothesis Allows

| Property of the returns                                | RW1 | RW2 | RW3        | Martingale |
|--------------------------------------------------------|-----|-----|------------|------------|
| $\text{Corr}(r_t, r_{t-k}) = 0$                        | yes | yes | yes        | yes        |
| $E[r_t \mid \text{past}] = \mu$ (unpredictable mean)   | yes | yes | not always | yes        |
| constant variance                                      | yes | no  | no         | no         |
| volatility clustering allowed                          | no  | no  | yes        | yes        |
| GARCH returns ( <a href="#">Chapter 9</a> ) satisfy it | no  | no  | yes        | yes        |

- **GARCH** (generalised autoregressive conditional heteroskedasticity): the variance of tomorrow depends on today's shocks; the mean stays unpredictable
- Weak-form efficiency needs only a martingale (or RW3), not RW1

## Martingale and Random Walk

- $\{p_t - \mu t\}$  is a **martingale** if  $E[p_{t+1} - \mu(t+1) \mid \mathcal{F}_t] = p_t - \mu t$  Samuelson (1965)
  - ▶ equivalently,  $E[r_{t+1} - \mu \mid \mathcal{F}_t] = 0$ : the returns minus  $\mu$  are a martingale difference
  - ▶ a martingale difference with finite variance is uncorrelated: martingale  $\Rightarrow$  RW3
- RW1 and RW2 imply the martingale; the converse fails: a martingale may have predictable variance
- **Question for the room:** does volatility clustering contradict weak-form efficiency?
  - ▶ **Answer:** no; it rejects RW1 and RW2, but a predictable variance gives no predictable return; the martingale and RW3 survive

## Worked Example: How Risk Grows with the Horizon

- S&P 500 since 1990: daily standard deviation  $\hat{\sigma} = 1.13\%$ 
  - ▶ under a random walk,  $\text{sd}(r_t(q)) = \sigma\sqrt{q}$
- Step by step
  - ▶ one week,  $q = 5$ :  $1.13 \times \sqrt{5} = 1.13 \times 2.24 = 2.53\%$
  - ▶ one month,  $q = 20$ :  $1.13 \times 4.47 = 5.07\%$
  - ▶ one year,  $q = 250$ :  $1.13 \times 15.81 = 17.92\%$
- This is the “square-root-of-time rule” used to scale VaR and volatility
- If returns are autocorrelated, the rule fails: the variance-ratio test of Section 7 measures by how much

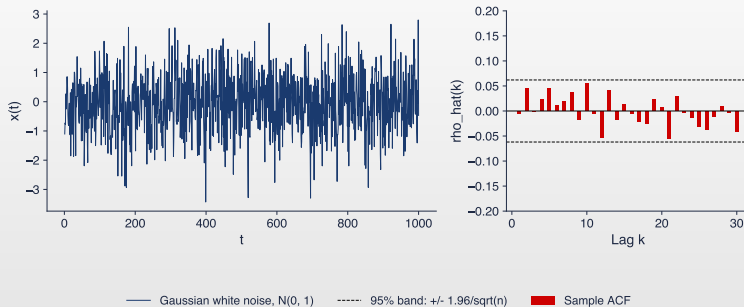
## Recap: Random Walk Hypotheses

- ▣  $p_t = \mu + p_{t-1} + \varepsilon_t$ : permanent shocks,  $\text{Var}(r_t(q)) = q\sigma^2$
- ▣ RW1 (i.i.d.)  $\Rightarrow$  RW2 (independent)  $\Rightarrow$  RW3 (uncorrelated)
- ▣ Martingale: unpredictable mean, possibly predictable variance; the right null for efficiency

## White Noise and Autocorrelation

- **Autocovariance** at lag  $k$ :  $\gamma(k) = \text{Cov}(x_t, x_{t-k})$ ; **autocorrelation**:  $\rho(k) = \gamma(k)/\gamma(0)$ 
  - ▶ the **ACF** (autocorrelation function):  $k \mapsto \rho(k)$ ,  $k = 1, 2, \dots$
- **White noise**: mean 0, constant variance  $\sigma^2$ ,  $\rho(k) = 0$  for every  $k \neq 0$ 
  - ▶ i.i.d. noise is white noise; Gaussian white noise is i.i.d.  $N(0, \sigma^2)$
  - ▶ white noise need not be independent: GARCH returns are white noise but not i.i.d.
- **Sample ACF**:  $\hat{\rho}(k) = \sum_{t=k+1}^T (x_t - \bar{x})(x_{t-k} - \bar{x}) / \sum_{t=1}^T (x_t - \bar{x})^2$ 
  - ▶ for i.i.d. data,  $\hat{\rho}(k) \approx N(0, 1/T)$ : 95% band  $\pm 1.96/\sqrt{T}$

## Gaussian White Noise and Its Sample ACF

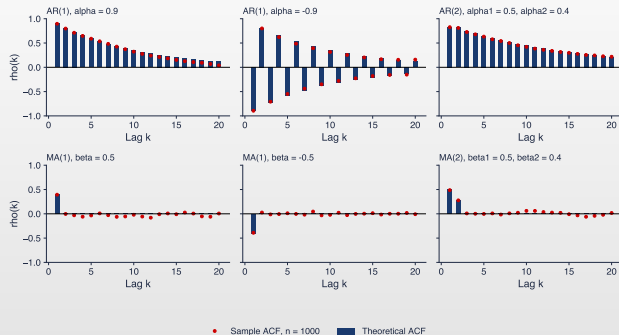


- Left: 1000 draws from  $N(0, 1)$  (as in SFetimewn); right:  $\hat{\rho}(1), \dots, \hat{\rho}(30)$  with the band  $\pm 1.96/\sqrt{1000} = \pm 0.062$
- Here 0 of 30 autocorrelations fall outside the band; with 30 lags we expect about 1.5 by chance alone

## The ACF of AR and MA Processes

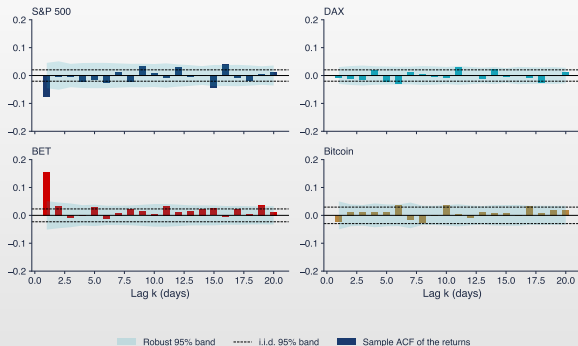
- ▣ **AR(1)** (autoregressive):  $x_t = \alpha x_{t-1} + \varepsilon_t$ ,  $|\alpha| < 1$ :  $\rho(k) = \alpha^k$ 
  - ▶ geometric decay; alternating signs if  $\alpha < 0$ ; example:  $\alpha = 0.9 \Rightarrow \rho(3) = 0.9^3 = 0.729$
- ▣ **AR(2)**:  $x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \varepsilon_t$ :  $\rho(1) = \alpha_1 / (1 - \alpha_2)$ ,  $\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2)$ 
  - ▶  $\alpha_1 = 0.5$ ,  $\alpha_2 = 0.4$ :  $\rho(1) = 0.5/0.6 = 0.833$ ,  $\rho(2) = 0.5 \times 0.833 + 0.4 = 0.817$
- ▣ **MA(1)** (moving average):  $x_t = \varepsilon_t + \beta \varepsilon_{t-1}$ :  $\rho(1) = \beta / (1 + \beta^2)$ ,  $\rho(k) = 0$  for  $k \geq 2$ 
  - ▶  $\beta = 0.5$ :  $\rho(1) = 0.5/1.25 = 0.40$ ; an MA( $q$ ) has an ACF that cuts off after lag  $q$
- ▣ The shape of the ACF tells the type of dependence: slow decay (AR), cut-off (MA), nothing (white noise)

## Theoretical and Sample ACF of Six Processes



- Bars: theoretical  $\rho(k)$ ; points: sample ACF of one simulated path of 1000 values (SFEacf1, SFEacf2, SFEacfma1, SFEacfma2)
- A random walk is the limit  $\alpha \rightarrow 1$  of the AR(1): its ACF would not decay at all

## The ACF of Daily Returns



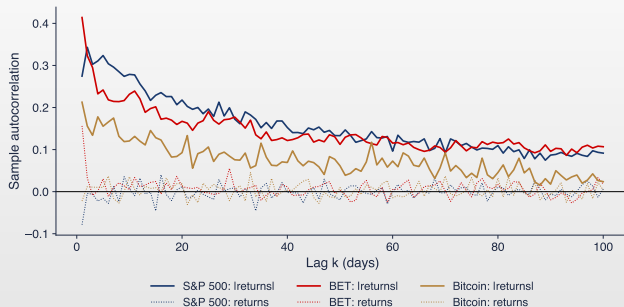
□ Bars:  $\hat{\rho}(1), \dots, \hat{\rho}(20)$ ; dashed: the i.i.d. band  $\pm 1.96/\sqrt{T}$ ; shaded: the robust band (next slide)

## Reading the ACF of Returns

- ▣ First autocorrelations: S&P 500  $-0.079$ , DAX  $-0.009$ , BET  $0.156$ , Bitcoin  $-0.022$ 
  - ▶ only the S&P 500 and the BET stand out at lag 1, with opposite signs
- ▣ The **robust** band:  $\pm 1.96 \sqrt{\hat{\delta}(k)/T}$ ,  $\hat{\delta}(k) = T \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$ ,  $e_t = r_t - \bar{r}$  Lo & MacKinlay (1988)
  - ▶ i.i.d. returns:  $\hat{\delta}(k) \approx 1$ ; volatility clustering: large returns follow large returns, so  $\hat{\delta}(k) > 1$
  - ▶ S&P 500 at lag 1:  $\pm 0.046$  instead of  $\pm 0.020$
- ▣ S&P 500, lags outside the band out of 20: 9 with the i.i.d. band, 3 with the robust band
- ▣ The i.i.d. band tests RW1; the robust band tests RW3, the hypothesis that matters for efficiency

 SFM\_ch7\_autocorrelation\_tests

## Returns Are Almost Uncorrelated, Their Size Is Not



- ACF of  $|r_t|$ : S&P 500 0.27 at lag 1, 0.22 at lag 20, 0.09 at lag 100; the returns themselves: dotted lines near 0
- Large moves follow large moves: RW1 and RW2 are rejected by eye; RW3 and the martingale are not touched by this chart

## Recap: White Noise and the ACF

- White noise: uncorrelated; i.i.d. noise: independent; GARCH returns: white noise, not i.i.d.
- AR: geometric decay of the ACF; MA( $q$ ): cut-off after  $q$  lags
- Use the robust band  $\pm 1.96\sqrt{\hat{\delta}(k)/T}$  for returns: the i.i.d. band is too narrow

## Portmanteau Tests: Box–Pierce and Ljung–Box

- $H_0: \rho(1) = \dots = \rho(m) = 0$ ;  $H_1$ : at least one  $\rho(k) \neq 0$ ; one test for  $m$  lags together
  - ▶ “portmanteau”: a suitcase that carries many autocorrelations at once
- Box–Pierce [Box & Pierce \(1970\)](#):  $Q_{BP}(m) = T \sum_{k=1}^m \hat{\rho}(k)^2$
- **LB** (Ljung–Box) [Ljung & Box \(1978\)](#):  $Q_{LB}(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}(k)^2}{T-k}$ , closer to  $\chi^2$  in small samples
  - ▶ under i.i.d. returns both  $\sim \chi^2(m)$ ; reject  $H_0$  if  $Q > \chi_{0.95}^2(m)$ ;  $\chi_{0.95}^2(10) = 18.31$
- Choice of  $m$ : here  $m = 10$  (two trading weeks); a larger  $m$  dilutes a short-lived effect

## Worked Example: Ljung–Box Step by Step

- ▣  $T = 1000$  daily returns,  $\hat{\rho}(1) = 0.06$ ,  $\hat{\rho}(2) = -0.04$ ,  $\hat{\rho}(3) = 0.03$ ; test with  $m = 3$ 
  - ▶  $T\hat{\rho}(k)^2$ : 3.6, 1.6, 0.9; their sum:  $Q_{BP}(3) = 6.10$
  - ▶  $Q_{LB}(3) = 1000 \times 1002 \times (0.06^2/999 + 0.04^2/998 + 0.03^2/997) = 6.12$
- ▣ Decision:  $\chi_{0.95}^2(3) = 7.81 > 6.12$ : do not reject  $H_0$  at 5%
  - ▶ note:  $\hat{\rho}(1) = 0.06$  lies just inside the band  $\pm 1.96/\sqrt{1000} = \pm 0.062$ ; the joint test does not reject either
- ▣ The same numbers with  $T = 5000$ :  $Q$  five times larger, a clear rejection: small autocorrelations become significant in long samples

## A Portmanteau Test Robust to Volatility Clustering

- The  $\chi^2$  law of  $Q_{LB}$  needs  $\text{Var}(\hat{\rho}(k)) = 1/T$ : true for i.i.d. returns, false with volatility clustering
  - ▶ then  $Q_{LB}$  rejects a true martingale far too often
- **Robust** statistic: divide each  $\hat{\rho}(k)^2$  by its own variance  $\hat{\delta}(k)/T$  Escanciano & Lobato (2009)
  - ▶  $\tilde{Q}(m) = \sum_{k=1}^m \frac{T\hat{\rho}(k)^2}{\hat{\delta}(k)} \sim \chi^2(m)$  for a martingale difference with finite fourth moment
  - ▶ with i.i.d. returns  $\hat{\delta}(k) \approx 1$  and  $\tilde{Q} \approx Q_{BP}$
- Rule: test RW1 with  $Q_{LB}$ ; test the martingale (weak-form efficiency) with  $\tilde{Q}$

## Autocorrelation Tests on Real Data

|         | $T$   | $\hat{\rho}(1)$ | $Q_{LB}(10)$ | $p$     | $\tilde{Q}(10)$ | $p$     | $Q_{LB}(10), r^2$ | runs $z$ | $p$     |
|---------|-------|-----------------|--------------|---------|-----------------|---------|-------------------|----------|---------|
| S&P 500 | 9 240 | -0.079          | 90.8         | < 0.001 | 18.8            | 0.043   | 8015              | 2.97     | 0.003   |
| DAX     | 9 288 | -0.009          | 21.8         | 0.016   | 8.6             | 0.571   | 3747              | 1.73     | 0.084   |
| BET     | 7 243 | 0.156           | 198.3        | < 0.001 | 42.9            | < 0.001 | 2500              | -7.75    | < 0.001 |
| Bitcoin | 4 384 | -0.022          | 20.2         | 0.028   | 11.9            | 0.294   | 213               | 2.68     | 0.007   |
| TLV     | 3 788 | 0.010           | 29.1         | 0.001   | 13.2            | 0.210   | 481               | -1.68    | 0.093   |
| SNP     | 3 815 | -0.015          | 20.7         | 0.023   | 9.9             | 0.446   | 505               | 1.72     | 0.085   |
| BRD     | 3 771 | 0.032           | 19.9         | 0.030   | 10.8            | 0.375   | 416               | -3.04    | 0.002   |
| TGN     | 3 725 | 0.055           | 28.7         | 0.001   | 12.3            | 0.263   | 639               | 1.49     | 0.136   |

- Classic LB: all eight series reject at 5%; robust  $\tilde{Q}$ : only the S&P 500 ( $p = 0.043$ ) and the BET ( $p = < 0.001$ )
- Squared returns: huge  $Q_{LB}$  everywhere: RW1 and RW2 are rejected for every series (volatility clustering)
- Start of the series: S&P 500 and DAX 1990; BET 1997; Bitcoin 2014; BVB stocks 2010 (TLV without 30–31 May 2016, a data error, [Chapter 2](#))

## The Runs Test

- ▣ A **run**: a sequence of returns with the same sign;  $++-++++--$  has 4 runs Wald & Wolfowitz (1940)
  - ▶ the test uses only the signs: it is **nonparametric**, insensitive to heavy tails and to volatility
- ▣ With  $n_1$  positive and  $n_2$  negative returns,  $n = n_1 + n_2$ , under independence:
  - ▶  $E[R] = \frac{2n_1n_2}{n} + 1$ ,  $\text{Var}(R) = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}$ ,  $z = \frac{R - E[R]}{\sqrt{\text{Var}(R)}} \approx N(0, 1)$
- ▣ Too few runs ( $z < 0$ ): signs persist, positive dependence; too many runs ( $z > 0$ ): signs alternate
  - ▶ BET: 3 278 runs against 3 606 expected,  $z = -7.75$ ; S&P 500: 4 738 against 4 596,  $z = 2.97$
  - ▶ the same directions as  $\hat{\rho}(1)$ : continuation in the BET, reversal in the S&P 500

 SFM\_ch7\_autocorrelation\_tests

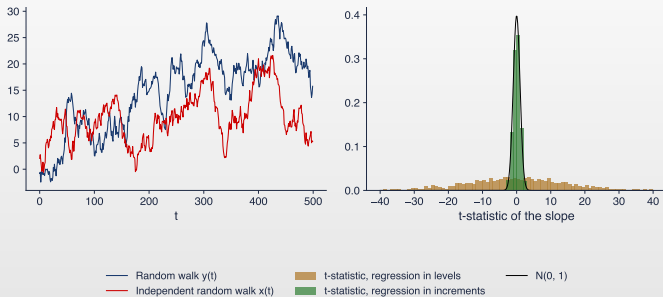
## Recap: Testing for Autocorrelation

- ▣  $Q_{LB}(m) = T(T+2) \sum \hat{\rho}(k)^2 / (T-k) \sim \chi^2(m)$  under i.i.d. returns
- ▣ With volatility clustering use the robust  $\tilde{Q}$ : the verdict changes for six of eight series
- ▣ Runs test: signs only; too few runs means continuation
- ▣  $\tilde{Q}$ : reversal in the S&P 500, continuation in the BET; no rejection for the DAX, Bitcoin and the BVB stocks

## Stationary and Integrated Series

- **(Weakly) stationary** series: constant mean, constant variance,  $\text{Cov}(x_t, x_{t-k})$  depends only on  $k$ 
  - ▶ shocks fade away; the series returns to its mean; notation  $I(0)$
- **Integrated of order 1**,  $I(1)$ : not stationary, but its first difference  $\Delta x_t = x_t - x_{t-1}$  is
  - ▶ the random walk is  $I(1)$ :  $p_t$  wanders,  $\Delta p_t = r_t$  is stationary
- **Unit root**: in  $p_t = \phi p_{t-1} + \varepsilon_t$  the case  $\phi = 1$ ;  $|\phi| < 1$  gives a stationary AR(1)
  - ▶ the name: the root of  $1 - \phi z = 0$  is  $z = 1/\phi = 1$
- Question of this section: are log prices  $I(1)$  and returns  $I(0)$ ?

## Why It Matters: Spurious Regression



- Regress one random walk on another, independent one ( $T = 500$ , 2 000 repetitions) [Granger & Newbold \(1974\)](#):  $|t| > 1.96$  in 90% of cases, median  $R^2 = 0.17$
- On the increments:  $|t| > 1.96$  in 5.5% of cases, as it should be; lesson: test for unit roots before regressing levels ( $|t| > 40$  in 2.6% of cases, not drawn)

## The Dickey–Fuller Test

- Subtract  $p_{t-1}$ :  $\Delta p_t = c + \gamma p_{t-1} + \varepsilon_t$  with  $\gamma = \phi - 1$  [Dickey & Fuller \(1979\)](#)
  - ▶  $H_0$ :  $\gamma = 0$  (unit root);  $H_1$ :  $\gamma < 0$  (stationary); a one-sided test
  - ▶ statistic:  $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$ , the usual  $t$ -ratio of OLS (ordinary least squares)
- Under  $H_0$ ,  $\tau$  does **not** follow a  $t$  or Normal law: its critical values are more negative
  - ▶ 5% critical values [MacKinnon \(1996\)](#):  $-2.86$  with a constant,  $-3.41$  with a constant and a trend (Normal:  $-1.645$ )
- Three specifications: no constant; constant  $c$ ; constant and trend  $c + bt$ 
  - ▶ log prices drift upwards: use constant and trend; returns: constant only

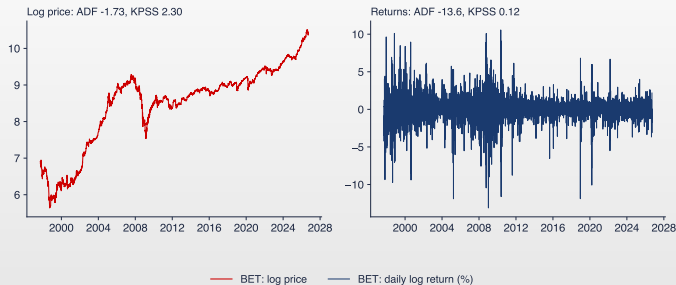
## The Augmented Test and a Worked Example

- ▣ **ADF** (augmented Dickey–Fuller) [Said & Dickey \(1984\)](#): add lagged differences to remove autocorrelation from  $\varepsilon_t$ 
  - ▶  $\Delta p_t = c + bt + \gamma p_{t-1} + \sum_{j=1}^L \delta_j \Delta p_{t-j} + \varepsilon_t$ ; same  $\tau$ , same critical values
  - ▶  $L$  chosen by AIC (Akaike information criterion, [Chapter 6](#)), at most  $12(T/100)^{1/4}$
- ▣ **Worked example**: BET log price, constant, no lags,  $T = 7\,243$ 
  - ▶ OLS:  $\hat{\gamma} = -0.00001$ ,  $\text{SE}(\hat{\gamma}) = 0.00016$
  - ▶  $\tau = -0.00001/0.00016 = -0.09 > -2.86$ : do not reject the unit root
  - ▶ a usual  $t$ -test (critical value  $-1.645$ ) would have rejected it: the wrong distribution gives the wrong answer

## Phillips–Perron and KPSS

- **PP** (Phillips–Perron) [Phillips & Perron \(1988\)](#): the Dickey–Fuller regression without lags;  $\tau$  is corrected for autocorrelation and heteroskedasticity
  - ▶ the correction uses a long-run variance with Bartlett weights [Newey & West \(1987\)](#); same  $H_0$  and critical values as ADF
- **KPSS** (Kwiatkowski–Phillips–Schmidt–Shin) [Kwiatkowski et al. \(1992\)](#): the hypotheses are **reversed**
  - ▶  $H_0$ : stationary (around a constant or a trend);  $H_1$ : unit root
  - ▶  $KPSS = \sum_{t=1}^T S_t^2 / (T^2 \hat{\lambda}^2)$ ,  $S_t$  the partial sums of the residuals,  $\hat{\lambda}^2$  their long-run variance
  - ▶ reject stationarity if  $KPSS > 0.463$  (constant) or  $> 0.146$  (trend), at 5%
- Use both: ADF/PP ask “is there evidence against a unit root?”; KPSS asks “is there evidence against stationarity?”
  - ▶ ADF does not reject and KPSS rejects:  $I(1)$ ; ADF rejects and KPSS does not:  $I(0)$ ; both or neither reject: inconclusive

## The BET: Log Price and Returns



- Log price: ADF  $-1.73$  ( $p = 0.736$ ), KPSS  $2.30$  ( $p < 0.01$ ): a unit root, not stationary
- Returns: ADF  $-13.65$  ( $p < 0.001$ ), KPSS  $0.12$  ( $p > 0.10$ ): stationary
- The 2008 fall looks like a temporary deviation, but the tests say: permanent shocks, no return to a trend line

## Unit-Root Tests on Prices and Returns

|         | log price (constant, trend) |       |      | returns (constant) |         |      |
|---------|-----------------------------|-------|------|--------------------|---------|------|
|         | ADF                         | PP    | KPSS | ADF                | PP      | KPSS |
| S&P 500 | -1.47                       | -1.57 | 2.23 | -17.61             | -105.05 | 0.12 |
| DAX     | -2.69                       | -2.65 | 0.94 | -23.30             | -97.40  | 0.04 |
| BET     | -1.73                       | -1.54 | 2.30 | -13.65             | -74.47  | 0.12 |
| Bitcoin | -1.48                       | -1.54 | 1.81 | -20.16             | -67.84  | 0.13 |
| TLV     | -3.46                       | -3.35 | 0.57 | -12.96             | -60.79  | 0.23 |
| SNP     | -0.80                       | -0.74 | 2.50 | -19.50             | -62.71  | 0.29 |
| BRD     | -2.18                       | -2.19 | 1.64 | -59.46             | -59.53  | 0.29 |
| TGN     | -0.96                       | -0.92 | 0.71 | -39.90             | -57.83  | 0.22 |

- 5% critical values: ADF and PP -3.41 (trend), -2.86 (constant); KPSS 0.146 (trend), 0.463 (constant)
- Log prices: no ADF/PP rejection except TLV (ADF  $p = 0.044$ ); KPSS rejects stationarity for all: prices are  $I(1)$
- Returns: ADF and PP reject strongly; KPSS never rejects ( $p > 0.10$ ): returns are  $I(0)$

## A Unit Root Is Not Proof of Efficiency

- ▣ A unit root in prices is **necessary** for a random walk, but **not sufficient**
  - ▶ example:  $r_t = 0.3 r_{t-1} + \varepsilon_t$  makes  $p_t$  an  $I(1)$  series whose increments are predictable
  - ▶ ADF cannot distinguish it from a random walk; the BET has a unit root and autocorrelated returns
- ▣ A stationary price (no unit root) would contradict efficiency: prices would return to a predictable level
  - ▶ TLV: ADF rejects at 5% with a trend, KPSS also rejects: an inconclusive case, not a profitable rule
- ▣ Unit-root tests have low power against  $\phi$  close to 1: the efficiency question is about the increments, and needs tests on returns

## Recap: Unit Roots

- ▣  $I(1)$ : shocks are permanent;  $I(0)$ : shocks fade; random walk =  $I(1)$  prices,  $I(0)$  returns
- ▣ ADF and PP:  $H_0$  unit root, Dickey–Fuller critical values; KPSS:  $H_0$  stationarity
- ▣ Our data: log prices  $I(1)$ , returns  $I(0)$ ; regressions in levels can be spurious
- ▣ A unit root is necessary, not sufficient, for a random walk

## The Idea of the Variance Ratio

- Under a random walk,  $\text{Var}(r_t(q)) = q \text{Var}(r_t)$ ; the **VR** (variance ratio) compares the two sides
  - ▶  $\text{VR}(q) = \frac{\text{Var}(r_t(q))}{q \text{Var}(r_t)} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho(k)$  Lo & MacKinlay (1988), Campbell, Lo & MacKinlay (1997)
- Reading the number
  - ▶  $\text{VR}(q) = 1$ : random walk (RW1, RW2, RW3 all give 1)
  - ▶  $\text{VR}(q) > 1$ : positive autocorrelation, **momentum** (moves continue)
  - ▶  $\text{VR}(q) < 1$ : negative autocorrelation, **mean reversion** (moves are partly undone)
- One number summarises  $q - 1$  autocorrelations with declining weights: more power against persistent, small autocorrelations

## Worked Example: VR from the ACF of the S&P 500

- $\hat{\rho}(1), \dots, \hat{\rho}(4)$  of the S&P 500:  $-0.079, -0.004, -0.006, -0.022$ 
  - ▶  $q = 2$ :  $\text{VR}(2) = 1 + \hat{\rho}(1) = 0.921$
  - ▶  $q = 5$ : weights  $2(1 - k/5) = 1.6, 1.2, 0.8, 0.4$ :  $\text{VR}(5) = 1 + 1.6\hat{\rho}(1) + 1.2\hat{\rho}(2) + 0.8\hat{\rho}(3) + 0.4\hat{\rho}(4) = 0.855$
- The Lo–MacKinlay estimator on the same data:  $\text{VR}(2) = 0.922, \text{VR}(5) = 0.856$ 
  - ▶ almost the same numbers: the two forms differ only by small-sample corrections
- Meaning: the weekly variance is about 0.856 of five daily variances: daily moves of the S&P 500 are partly reversed within a week

## Lo and MacKinlay (1988): Estimator and Tests

- With  $T$  returns, mean  $\hat{\mu}$  and overlapping  $q$ -day sums Lo & MacKinlay (1988):

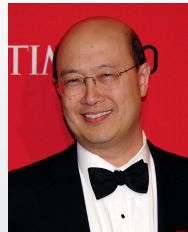
- $\hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t (r_t - \hat{\mu})^2$ ,  $\hat{\sigma}_c^2(q) = \frac{1}{m} \sum_t (r_t(q) - q\hat{\mu})^2$

- $m = q(T - q + 1)(1 - q/T)$  corrects the bias;  $\widehat{VR}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$

- Homoskedastic test (RW1):**  $Z(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0, 1)$

- Robust test (RW3, martingale):**  $Z^*(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{\hat{\theta}(q)/T}} \approx N(0, 1)$

- $\hat{\theta}(q) = \sum_{j=1}^{q-1} [2(q-j)/q]^2 \hat{\delta}(j)$ , with the  $\hat{\delta}(j)$  of the robust ACF band

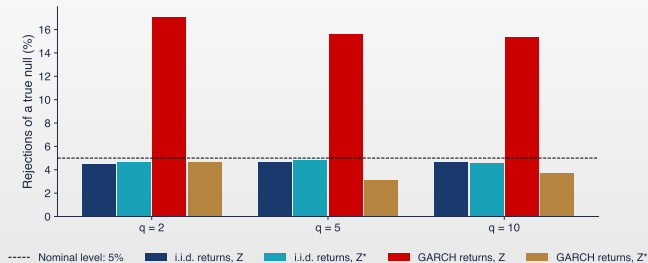


Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0;

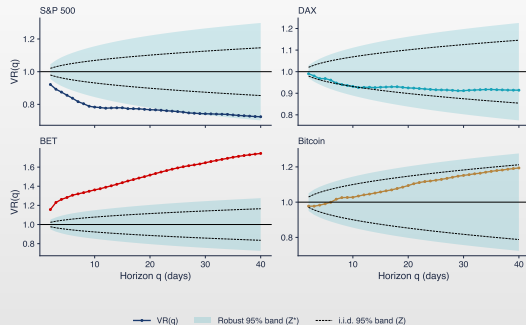
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 SFM\_ch7\_variance\_ratio

## Why the Robust Test: a Monte Carlo Check



- 1 000 samples of  $T = 2000$  returns with  $H_0$  true: i.i.d. Normal, and GARCH(1,1) ( $\alpha = 0.12$ ,  $\beta = 0.86$ ): uncorrelated, with volatility clustering
- i.i.d.: both reject about 5%; GARCH:  $Z(2)$  rejects 17.1% of the time,  $Z^*(2)$  only 4.7%
- The size of a test: how often it rejects a true  $H_0$ ;  $Z$  finds “inefficiency” that is only volatility clustering

## Variance Ratios for Horizons of 2 to 40 Days



- S&P 500: VR falls below 1 at every horizon (mean reversion); BET: VR rises up to about 1.517 at 20 days and beyond (momentum)
- DAX and Bitcoin: inside the robust band (shaded) at every horizon; the i.i.d. band (dashed) is much narrower

## Variance Ratios and Robust Tests

|         | VR(2) | VR(5) | VR(10) | VR(20) | $Z^*(2)$ | $Z^*(5)$ | $Z^*(10)$ | $Z^*(20)$ |
|---------|-------|-------|--------|--------|----------|----------|-----------|-----------|
| S&P 500 | 0.922 | 0.856 | 0.783  | 0.767  | -3.37    | -2.76    | -2.73     | -2.06     |
| DAX     | 0.990 | 0.968 | 0.932  | 0.925  | -0.61    | -0.88    | -1.21     | -0.91     |
| BET     | 1.156 | 1.283 | 1.362  | 1.517  | 5.99     | 5.25     | 4.74      | 4.94      |
| Bitcoin | 0.977 | 0.991 | 1.027  | 1.094  | -0.88    | -0.19    | 0.37      | 0.91      |
| TLV     | 1.009 | 0.997 | 0.954  | 0.995  | 0.34     | -0.04    | -0.48     | -0.04     |
| SNP     | 0.985 | 0.960 | 0.975  | 1.038  | -0.53    | -0.68    | -0.29     | 0.32      |
| BRD     | 1.032 | 1.083 | 1.078  | 1.107  | 1.28     | 1.55     | 0.97      | 0.96      |
| TGN     | 1.055 | 1.150 | 1.142  | 1.166  | 2.05     | 2.60     | 1.69      | 1.44      |

- $|Z^*| > 1.96$ : reject the random walk at 5% for that horizon
- S&P 500: significant mean reversion at all four horizons; BET: significant momentum at all four
- Transgaz:  $Z^*(2) = 2.05$  and  $Z^*(5) = 2.60$ ; DAX, Bitcoin, TLV, SNP, BRD: no rejection
- But eight tests per series at once: some “significant” horizons appear by chance

## Several Horizons at Once: the Chow–Denning Test

- ▣ Testing  $q = 2, 5, 10, 20$  separately at 5% gives a much higher chance of at least one false rejection
  - ▶ for 4 independent tests:  $1 - 0.95^4 = 18.5\%$
- ▣ **CD** (Chow–Denning) statistic [Chow & Denning \(1993\)](#):  $CD = \max_i |Z^*(q_i)|$ ,  $i = 1, \dots, m$ 
  - ▶ critical value from the studentised maximum modulus law:  $P(CD \leq c) = (2\Phi(c) - 1)^m$
  - ▶  $m = 4$ , 5%:  $c = 2.49$  instead of 1.96;  $\Phi$ : the standard Normal distribution function
- ▣ Reject the random walk if the largest  $|Z^*|$  exceeds 2.49: one decision for all horizons

## An Automatic Choice of the Horizon

- The horizon  $q$  can be chosen from the data: the automatic VR test [Choi \(1999\)](#)
  - ▶  $VR(k) = 1 + 2 \sum_{i=1}^{T-1} w(i/k) \hat{\rho}(i)$ , with smoothly declining weights  $w$  (quadratic spectral kernel) [Andrews \(1991\)](#)
  - ▶ the bandwidth  $k$  grows with the first-order autocorrelation and with  $T$ ; statistic  $\sqrt{T/k} (VR(k) - 1) / \sqrt{2} \approx N(0, 1)$
- **Wild bootstrap** p-value [Kim \(2009\)](#): resample  $r_t^* = \eta_t(r_t - \bar{r})$  with  $\eta_t \sim N(0, 1)$ , 500 times
  - ▶ each  $r_t^*$  keeps the size of  $r_t$  (its volatility) but loses any autocorrelation: a robust null distribution
- Three tools, one question:  $Z^*(q)$  for a chosen horizon, CD for a set of horizons, the automatic test for a horizon chosen by the data

## Joint and Automatic Tests on Real Data

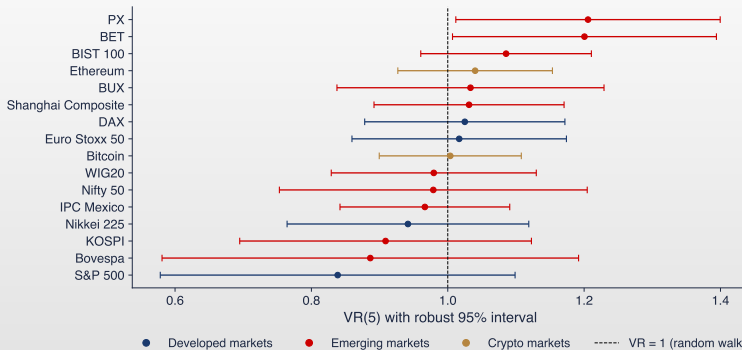
|         | $T$   | CD   | $p$     | $\hat{k}$ | automatic VR | $p$ (bootstrap) |
|---------|-------|------|---------|-----------|--------------|-----------------|
| S&P 500 | 9 240 | 3.37 | 0.003   | 3.7       | -5.21        | 0.006           |
| DAX     | 9 288 | 1.21 | 0.644   | 1.7       | -0.39        | 0.608           |
| BET     | 7 243 | 5.99 | < 0.001 | 5.6       | 8.86         | < 0.001         |
| Bitcoin | 4 384 | 0.91 | 0.835   | 2.0       | -1.01        | 0.352           |
| TLV     | 3 788 | 0.48 | 0.981   | 1.5       | 0.31         | 0.718           |
| SNP     | 3 815 | 0.68 | 0.937   | 1.7       | -0.53        | 0.562           |
| BRD     | 3 771 | 1.55 | 0.403   | 2.3       | 1.58         | 0.152           |
| TGN     | 3 725 | 2.60 | 0.037   | 3.0       | 3.78         | 0.004           |

- Both tests reject for the S&P 500, the BET and Transgaz; neither rejects for DAX, Bitcoin, TLV, SNP and BRD
- $\hat{k}$  is large where the first autocorrelation is large (BET 5.6), small where it is near 0 (DAX 1.7)
- Interpretation: rejection does not mean profit: an autocorrelation of  $-0.079$  predicts less than 1% of the variance of tomorrow's return; trading costs absorb most of it

## Recap: The Variance-Ratio Test

- ▣  $VR(q) = 1 + 2 \sum_{k < q} (1 - k/q) \rho(k)$ ;  $> 1$  momentum,  $< 1$  mean reversion
- ▣ Use  $Z^*(q)$ : the homoskedastic  $Z(q)$  over-rejects under volatility clustering
- ▣ Several horizons: Chow–Denning; data-driven horizon: automatic VR with wild bootstrap
- ▣ S&P 500 mean-reverting, BET trending; DAX, Bitcoin and most BVB stocks consistent with a random walk

## Sixteen Markets over the Last Ten Years



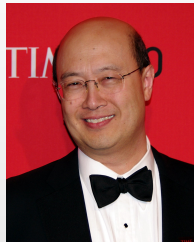
- VR(5) with its robust 95% interval, from 19 September 2016 to 18 September 2026: developed, emerging and crypto markets

## Reading the Cross-Market Comparison

- ▣ Only two intervals exclude 1: the PX ( $Z^*(5) = 2.08$ ) and the BET ( $Z^*(5) = 2.03$ ), both above 1
  - ▶ two smaller Central European markets, with less liquid index stocks
- ▣ **Many markets, many tests:** 16 tests at 5% give 0.8 false rejections on average
  - ▶ if the tests were independent, the chance of at least one false rejection would be  $1 - 0.95^{16} = 56\%$
  - ▶ Chow–Denning over four horizons: no market rejects at 5% (BET  $p = 0.159$ , PX  $p = 0.142$ )
- ▣ S&P 500:  $VR(5) = 0.84$ , the lowest, but with a wide interval ( $Z^* = -1.22$ ): the large swings of 2020 inflate  $\hat{\theta}$
- ▣ Bitcoin and Ethereum:  $VR(5)$  close to 1, as for developed markets

## The Adaptive Markets Hypothesis

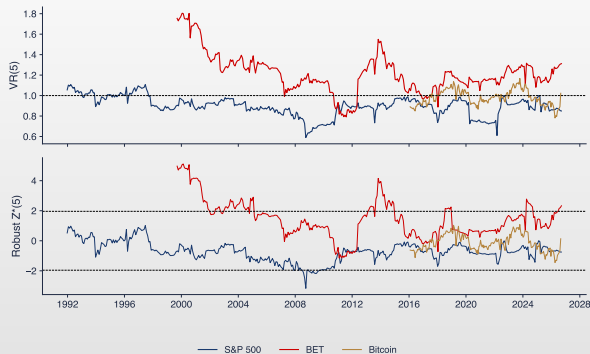
- **AMH** (adaptive markets hypothesis) [Lo \(2004\)](#): markets are ecosystems of traders who learn and adapt
  - ▶ profit opportunities appear, are exploited and disappear; efficiency changes with the environment
  - ▶ crises, new technology, new participants and regulation move the market away from efficiency or back towards it
- Testable consequence: predictability varies over time
  - ▶ evidence: [Kim, Shamsuddin & Lim \(2011\)](#) (U.S., a century of data); [Urquhart & McGroarty \(2016\)](#); survey: [Lim & Brooks \(2011\)](#)
- Tool: the same test on rolling windows



Andrew W. Lo, author of the AMH

Photo: [David Shankbone \(2012\)](#); CC BY 3.0;  
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## Rolling Variance Ratios



- $VR(5)$  and  $Z^*(5)$  on windows of 500 trading days (about two years), a new window every 21 days, dated at the window end

## Reading the Rolling Variance Ratios

- ▣ BET: VR(5) up to 1.80 in the window ending in 2000; closer to 1 after 2008, apart from short episodes
  - ▶ 29% of the windows reject at 5%, all of them with  $VR > 1$  (momentum)
- ▣ S&P 500: VR(5) mostly below 1; the lowest, 0.59, in the window ending in 2008
  - ▶ 4% of the windows reject, all with  $VR < 1$ : reversals are strongest in crises
- ▣ Bitcoin: no window rejects since 2016 (185 windows)
- ▣ Caution: consecutive windows share 479 of 500 days; the share of rejecting windows is a description, not a test

## Three Sub-Periods per Market

|         | period    | $T$   | $\hat{\rho}(1)$ | VR(5) | $Z^*(5)$ | CD p    |
|---------|-----------|-------|-----------------|-------|----------|---------|
| BET     | 1997–2006 | 2 306 | 0.270           | 1.50  | 6.15     | < 0.001 |
| BET     | 2007–2016 | 2 516 | 0.066           | 1.08  | 0.82     | 0.262   |
| BET     | 2017–2026 | 2 421 | 0.064           | 1.20  | 2.04     | 0.154   |
| S&P 500 | 1990–2001 | 3 025 | 0.017           | 0.96  | −0.62    | 0.334   |
| S&P 500 | 2002–2013 | 3 019 | −0.100          | 0.80  | −2.53    | 0.010   |
| S&P 500 | 2014–2026 | 3 196 | −0.124          | 0.85  | −1.30    | 0.080   |
| Bitcoin | 2014–2017 | 1 201 | 0.028           | 0.98  | −0.23    | 0.983   |
| Bitcoin | 2018–2021 | 1 461 | −0.058          | 0.98  | −0.21    | 0.374   |
| Bitcoin | 2022–2026 | 1 722 | −0.026          | 1.01  | 0.18     | 0.924   |

- BET: strong momentum in 1997–2006 ( $\hat{\rho}(1) = 0.270$ ), much weaker afterwards; since 2017 VR(5) is still above 1, but the joint test does not reject
- S&P 500: the negative autocorrelation appears after 2002; Bitcoin: no rejection in any period since 2014; [Urquhart \(2016\)](#) found inefficiency in its earliest years

## Why the Early BET Was Predictable

- **Thin trading:** few trades per day in many index stocks
  - ▶ a closing price may be hours old; news enters the index over several days
  - ▶ the index then shows positive autocorrelation even if each stock is efficient
- Changes after 2007
  - ▶ more liquidity, foreign investors, EU membership, the FTSE Russell upgrade to Secondary Emerging (2020) [FTSE Russell \(2019\)](#)
- Crashes on the Bucharest Stock Exchange: [Pele & Mazurencu-Marinescu \(2012\)](#)



The Stock Exchange Palace, Bucharest

Photo: [Neoclassicism Enthusiast \(2019\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

## Recap: Efficiency across Markets and over Time

- Over the last ten years most markets, including Bitcoin, are consistent with a random walk
- Efficiency changes over time: the AMH; rolling windows show it, but overlap
- The early BET: thin trading and momentum; today: much closer to a random walk
- Many markets, many tests: expect some false rejections

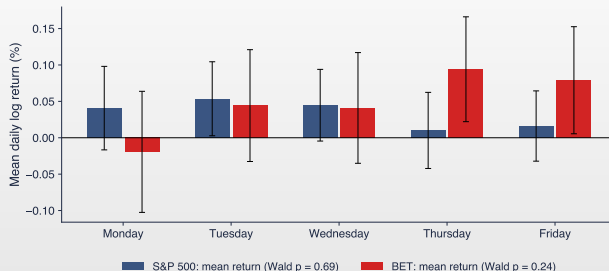
## Event Studies and the Strong Form

- **Event study** (semi-strong form) [MacKinlay \(1997\)](#), [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 4: how fast does a price absorb public news?
  - ▶ abnormal return: the return minus the return of a model (for example the market); cumulated over a window around the announcement
  - ▶ efficient: a jump on the day of the news, no drift afterwards
  - ▶ recent research: how fast futures prices absorb news, measured in nanoseconds at Eurex and CME [Tak, Schlamp, Osterrieder & Pele \(2026\)](#); market responses to Ethereum upgrades [Lin, Găman, Wang & Pele \(2025\)](#)
- **Strong form**: can anyone with private information earn excess returns?
  - ▶ insiders (managers trading their own shares) do: insider trading is illegal for this reason
  - ▶ professional fund managers mostly do not [Jensen \(1968\)](#), [Fama & French \(2010\)](#)

## Calendar Anomalies

- An **anomaly**: a pattern in returns that the equilibrium model does not explain
  - ▶ **weekend effect**: low or negative Monday returns French (1980)
  - ▶ **January effect**: higher returns in January, mostly for small stocks Rozeff & Kinney (1976)
  - ▶ **turn-of-the-month effect**: returns concentrated around the first trading days of the month Ariel (1987)
- Do anomalies last?
  - ▶ returns of published anomalies fall after publication: traders learn McLean & Pontiff (2016)
  - ▶ hundreds of patterns have been tested: a new one needs  $|t| > 3$ , not 2 Harvey, Liu & Zhu (2016)
- Data mining: with enough calendar splits, some will look significant by chance

## Day-of-Week Returns: S&P 500 and BET



- Mean daily return by weekday with 95% intervals (Newey–West standard errors [Newey & West \(1987\)](#)); Wald test of equal means: S&P 500  $p = 0.69$ , BET  $p = 0.24$
- BET Monday:  $-0.019\%$ , the only negative mean, yet not significant; January: BET  $0.185\%$  per day against  $0.036\%$  ( $t = 1.72$ ,  $p = 0.09$ ); S&P 500:  $p = 0.59$
- No calendar effect survives at 5% in our data

## Recap: Semi-Strong Form, Strong Form and Anomalies

- Event studies: prices react to public news at once, with no drift, in efficient markets
- Strong form fails for insiders, holds roughly for professional managers
- Calendar anomalies: weak in our data, and they fade after publication

## An Open Question

- **Did the BET become more efficient after Romania's upgrade to Secondary Emerging market (September 2020)?** [FTSE Russell \(2019\)](#)
  - ▶ five years before:  $\hat{\rho}(1) = -0.002$ ,  $VR(5) = 1.12$  ( $Z^* = 0.80$ ); five years after:  $\hat{\rho}(1) = 0.087$ ,  $VR(5) = 1.18$  ( $Z^* = 1.78$ )
  - ▶ difference of the two  $VR(5)$ :  $z = 0.34$ : no evidence of a change
- Why it is open: COVID-19, new large listings and higher volumes happened at the same time; is the index or each stock the right unit?
- AI tools can speed up such a study; they do not replace checking it [Wang et al. \(2023\)](#)

 [SFM\\_ch7\\_adaptive\\_markets](#)

## How AI Could Help

- ▣ **Literature:** list studies of efficiency in Central and Eastern European markets and the tests they use
- ▣ **Code:** a first draft of rolling Lo-MacKinlay and automatic VR tests for BET stocks, with wild-bootstrap p-values
- ▣ **Robustness:** other windows, other horizons, BET-TR instead of BET, trading volume as a control
- ▣ Example prompt
  - ▶ Write a Python function that computes the Lo-MacKinlay variance ratio  $VR(5)$  and the heteroskedasticity-robust z-statistic for daily BET log returns on rolling 500-day windows, and test whether the mean VR differs before and after 21 September 2020.

## What to Check

- ▣ The statistic: robust  $Z^*$ , not the homoskedastic  $Z$ ; a draft that uses  $Z$  will “find” inefficiency
- ▣ Log returns of the index, not price levels; no unit-root test on returns used as an efficiency test
- ▣ Rolling windows overlap: a  $t$ -test on the mean of rolling statistics treats dependent numbers as independent
- ▣ The event date and the window are fixed before looking at the result
- ▣ References: every cited paper must exist; check the DOI

## Project Idea

- **Question:** is the Bucharest market more efficient today than in 2015?
  - ▶ a second question: does the change appear in the index or in the individual stocks?
  - ▶ data: BET since 1997, BET-TR since 2014, BVB stocks (TLV, SNP, BRD, TGN) since 2010 (EODHD)
- **Steps**
  - ▶ rolling VR(2), VR(5) with  $Z^*$  and the Chow–Denning test for the index and for each stock
  - ▶ compare 2015–2020 with 2020–2025 by a block bootstrap of the difference
  - ▶ repeat for the WIG20 and the BUX as controls: did all of them change at the same time?
- **Deliverable:** one table, one chart, and a paragraph on what the data can and cannot show
- **Declare any AI use, and list the errors of the AI that you corrected**

## Key Takeaways

- ▣ Efficient market: prices reflect information; weak form = past prices cannot be used to earn excess returns
- ▣ The right null is the martingale (RW3), not RW1: volatility clustering is not inefficiency
- ▣ Log prices are  $I(1)$  and returns  $I(0)$  in all our series; a unit root alone does not prove efficiency
- ▣ Use robust tests:  $\tilde{Q}$  and  $Z^*(q)$ ; join horizons with Chow–Denning or let the data choose (automatic VR)
- ▣ Evidence: S&P 500 slightly mean-reverting, early BET trending, most markets close to efficient today; efficiency varies over time
- ▣ Statistical predictability is not profit: costs and risk decide

## Key Formulas

| Quantity       | Formula                                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Random walk    | $p_t = \mu + p_{t-1} + \varepsilon_t, \quad \text{Var}(r_t(q)) = q\sigma^2$                                                                  |
| Sample ACF     | $\hat{\rho}(k) = \sum_t e_t e_{t-k} / \sum_t e_t^2, \quad \text{band } \pm 1.96 \sqrt{\hat{\delta}(k)/T}$                                    |
| Ljung–Box      | $Q_{LB}(m) = T(T+2) \sum_{k=1}^m \hat{\rho}(k)^2 / (T-k) \sim \chi^2(m)$                                                                     |
| Runs test      | $E[R] = 2n_1 n_2 / n + 1, \quad z = (R - E[R]) / \text{sd}(R)$                                                                               |
| ADF            | $\Delta p_t = c + bt + \gamma p_{t-1} + \sum_j \delta_j \Delta p_{t-j} + \varepsilon_t, \quad \tau = \hat{\gamma} / \text{SE}(\hat{\gamma})$ |
| KPSS           | $\sum_t S_t^2 / (T^2 \hat{\lambda}^2), \quad H_0: \text{stationarity}$                                                                       |
| VR             | $\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho(k)$                                                                                    |
| Robust VR test | $Z^*(q) = (\widehat{\text{VR}}(q) - 1) / \sqrt{\hat{\theta}(q)/T}, \quad \hat{\theta}(q) = \sum_{j < q} [2(q-j)/q]^2 \hat{\delta}(j)$        |
| Chow–Denning   | $\max_i  Z^*(q_i) , \quad P(\text{CD} \leq c) = (2\Phi(c) - 1)^m$                                                                            |

## Check Yourself

- **Question:**  $\hat{\rho}(1) = 0.10$  and  $\hat{\rho}(k) = 0$  for  $k \geq 2$ ; what does  $VR(2)$  tell us?
  - ▶ **Answer:**  $1 + 0.10 = 1.10$ : the two-day variance is 10% larger than under a random walk: momentum
- **Question:** ADF does not reject on log prices and KPSS rejects; is the market efficient?
  - ▶ **Answer:** we only know that prices are  $I(1)$ ; efficiency needs tests on the increments (ACF, VR)
- **Question:**  $Q_{LB}(10)$  rejects strongly for squared returns; is weak-form efficiency rejected?
  - ▶ **Answer:** no; this is volatility clustering; it rejects RW1 and RW2, not the martingale
- Next: [Chapter 8](#), volatility estimators and volatility clustering

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